Simulation and Modeling

Lecture 12

Output Data Analysis

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Simulation output of a Bank Model

TABLE 9.1
Results for 10 independent replications of the bank model

Replication	Number served	Finish time (hours)	Average delay in queue (minutes)	Average queue length	Proportion of customers delayed < 5 minutes
1	484	8.12	1.53	1.52	0.917
2	475	8.14	1.66	1.62	0.916
3	484	8.19	1.24	1.23	0.952
4	483	8.03	2.34	2.34	0.822
5	455	8.03	2.00	1.89	0.840
6	461	8.32	1.69	1.56	0.866
7	451	8.09	2.69	2.50	0.783
8	486	8.19	2.86	2.83	0.782
9	502	8.15	1.70	1.74	0.873
10	475	8.24	2.60	2.50	0.779

Statistical Analysis

EXAMPLE 9.14. For the bank of Example 9.1, suppose that we want to obtain a point estimate and an approximate 90 percent confidence interval for the expected average delay of a customer over a day, which is given by

$$E(X) = E\left(\frac{\sum_{i=1}^{N} D_i}{N}\right)$$

(Note that we estimate the expected *average* delay, since each delay has, in general, a different mean.) From the 10 replications given in Table 9.1 we obtained

$$\overline{X}(10) = 2.03, \qquad S^2(10) = 0.31$$

and

$$\overline{X}(10) \pm t_{9,0.95}\sqrt{\frac{S^2(10)}{10}} = 2.03 \pm 0.32$$

Thus, subject to the correct interpretation to be given to confidence intervals (see Sec. 4.5), we can claim with approximately 90 percent confidence that $E(X)$ is contained in the interval [1.71, 2.35] minutes.

Average Cost – Inventory System

EXAMPLE 9.15. For the inventory system of Sec. 1.5 and Example 9.8, suppose that we want to obtain a point estimate and an approximate 95 percent confidence interval for the expected average cost over the 120-month planning horizon, which is given by

$$E(X) = E\left(\frac{\sum_{i=1}^{120} C_i}{120}\right)$$

Average Cost – Inventory System

We made 10 independent replications and obtained the following X_j's:

129.35	127.11	124.03	122.13	120.44
118.39	130.17	129.77	125.52	133.75

which resulted in

$$\overline{X}(10) = 126.07, \qquad S^2(10) = 23.55$$

and the 95 percent confidence interval

$$126.07 \pm 3.47 \qquad \text{or, alternatively,} \qquad [122.60, 129.54]$$

Note that the estimated coefficient of variation (see Table 6.5), a measure of variability, is 0.04 for the inventory system and 0.27 for the bank model. Thus the X_j's for the bank model are inherently more variable than those for the inventory system.

Estimation of Expected Fraction

EXAMPLE 9.16. For the bank of Example 9.1, suppose that we would like to obtain a point estimate and an approximate 90 percent confidence interval for the expected proportion of customers with a delay less than 5 minutes over a day, which is given by

$$E(X) = E\left(\frac{\sum_{i=1}^{N} I_i(0,5)}{N}\right)$$

where the *indicator function* $I_i(0, 5)$ is defined as

$$I_i(0,5) = \begin{cases} 1 & \text{if } D_i < 5 \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, 2, \ldots, N$. From the last column of Table 9.1, we obtained

$$\bar{X}(10) = 0.853, \qquad S^2(10) = 0.004$$

and the 90 percent confidence interval

$$0.853 \pm 0.036 \qquad \text{or} \qquad [0.817, 0.889]$$

CI with Desired Interval

Suppose that we have constructed a confidence interval for μ based on a fixed number of replications n. If we assume that our estimate $S^2(n)$ of the population variance will not change (appreciably) as the number of replications increases, an *approximate* expression for the total number of replications, $n_a^*(\beta)$, required to obtain an absolute error of β is given by

$$n_a^*(\beta) = \min\left\{ i \geq n\colon t_{i-1,1-\alpha/2}\sqrt{\frac{S^2(n)}{i}} \leq \beta \right\} \tag{9.2}$$

Sometimes we are interested in a two-sided confidence interval of a certain level, say $1-\alpha$, and the problem is to choose the sample size n so that the interval is of a certain size. For instance, suppose that we want to compute an interval of length .1 that we can assert, with 99 percent confidence, contains μ. How large need n be? To solve this, note that as $z_{.005} = 2.58$ it follows that the 99 percent confidence interval for μ from a sample of size n is

$$\left(\overline{x} - 2.58\frac{\sigma}{\sqrt{n}}, \quad \overline{x} + 2.58\frac{\sigma}{\sqrt{n}}\right)$$

Hence, its length is

$$5.16\frac{\sigma}{\sqrt{n}}$$

Thus, to make the length of the interval equal to .1,

$$5.16\frac{\sigma}{\sqrt{n}} = .1$$

or

$$n = (51.6\,\sigma)^2$$

EXAMPLE 9.17. For the bank of Example 9.14, suppose that we would like to estimate the expected average delay with an absolute error of 0.25 minute and a confidence level of 90 percent. From the 10 available replications, we get

$$n_a^*(0.25) = \min\left\{ i \geq 10\colon t_{i-1,0.95}\sqrt{\frac{0.31}{i}} \leq 0.25 \right\} = 16$$

Metrics other than Averages

EXAMPLE 9.20. Consider the bank of Example 9.14, where the utilization factor $\rho = \lambda/(5\omega) = 0.8$. We compare the policy of having one queue for each teller (and jockeying) with the policy of having one queue feed all tellers on the basis of *expected average delay in queue* (see Example 9.14) and *expected time-average number of customers in queue*, which is defined by

$$E\left[\frac{\int_0^T Q(t)\,dt}{T}\right]$$

TABLE 9.4
Simulation results for the two bank policies: averages

	Estimates	
Measure of performance	**Five queues**	**One queue**
Expected operating time, hours	8.14	8.14
Expected average delay, minutes	5.57	5.57
Expected average number in queue	5.52	5.52

TABLE 9.5

Simulation results for the two bank policies: proportions

Interval (minutes)	Estimates of expected proportions of delays in interval — Five queues	Estimates of expected proportions of delays in interval — One queue
[0, 5)	0.626	0.597
[5, 10)	0.182	0.188
[10, 15)	0.076	0.107
[15, 20)	0.047	0.095
[20, 25)	0.031	0.013
[25, 30)	0.020	0
[30, 35)	0.015	0
[35, 40)	0.003	0
[40, 45)	0	0

Approx. CI for Bernoulli Random Variables

If we let X denote the number of the n items that meet the standards, then X is a binomial random variable with parameters n and p. Thus, when n is large, it follows by the normal approximation to the binomial that X is approximately normally distributed with mean np and variance $np(1-p)$. Hence,

$$\frac{X - np}{\sqrt{np(1-p)}} \dot{\sim} \mathcal{N}(0,1) \tag{7.5.1}$$

where $\dot{\sim}$ means "is approximately distributed as." Therefore, for any $\alpha \in (0,1)$,

$$P\left\{-z_{\alpha/2} < \frac{X - np}{\sqrt{np(1-p)}} < z_{\alpha/2}\right\} \approx 1 - \alpha$$

and so if X is observed to equal x, then an approximate $100(1-\alpha)$ percent confidence *region* for p is

$$\left\{p : -z_{\alpha/2} < \frac{x - np}{\sqrt{np(1-p)}} < z_{\alpha/2}\right\}$$

The foregoing region, however, is not an interval. To obtain a confidence *interval* for p, let $\hat{p} = X/n$ be the fraction of the items that meet the standards. From

Example 7.2.a, $\hat{p}$ is the maximum likelihood estimator of p, and so should be approximately equal to p. As a result, $\sqrt{n\hat{p}(1-\hat{p})}$ will be approximately equal to $\sqrt{np(1-p)}$ and so from Equation (7.5.1) we see that

$$\frac{X - np}{\sqrt{n\hat{p}(1-\hat{p})}} \dot{\sim} \mathcal{N}(0, 1)$$

Hence, for any $\alpha \in (0, 1)$ we have that

$$P\left\{-z_{\alpha/2} < \frac{X - np}{\sqrt{n\hat{p}(1-\hat{p})}} < z_{\alpha/2}\right\} \approx 1 - \alpha$$

or, equivalently,

$$P\{-z_{\alpha/2}\sqrt{n\hat{p}(1-\hat{p})} < np - X < z_{\alpha/2}\sqrt{n\hat{p}(1-\hat{p})}\} \approx 1 - \alpha$$

Dividing all sides of the preceding inequality by n, and using that $\hat{p} = X/n$, the preceding can be written as

$$P\{\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n} < p < \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\} \approx 1 - \alpha$$

which yields an approximate $100(1-\alpha)$ percent confidence interval for p.

Let X be a random variable defined on a replication as described in Sec. 9.4.1. Suppose that we would like to estimate the probability $p = P(X \in B)$, where B is a set of real numbers. {For example, B could be the interval $[20, \infty)$ in Example 9.20.} Make n independent replications and let $X_1, X_2, \ldots, X_n$ be the resulting IID random variables. Let S be the number of X_j's that fall in the set B. Then S has a binomial distribution (see Sec. 6.2.3) with parameters n and p, and an unbiased point estimator for p is given by

$$\hat{p} = \frac{S}{n}$$

Furthermore, if n is "sufficiently large," then an approximate $100(1 - \alpha)$ percent confidence interval for p is given by

$$\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

EXAMPLE 9.21. For the bank of Example 9.14, suppose that we would like to get a point estimate and approximate 90 percent confidence interval for

$$p = P(X \leq 15) \qquad \text{where } X = \max_{0 \leq t \leq T} Q(t)$$

In this case, $B = [0, 15]$. We made 100 independent replications of the bank simulation and obtained $\hat{p} = 0.77$. Thus, for approximately 77 out of every 100 days, we expect the maximum queue length during a day to be less than or equal to 15 customers. We also obtained the following approximate 90 percent confidence interval for p:

$$0.77 \pm 0.07 \qquad \text{or, alternatively,} \qquad [0.70, 0.84]$$

Estimate of the q-quantile

Suppose now that we would like to estimate the q-quantile (100qth percentile) x_q of the distribution of the random variable X (see Sec. 6.4.3 for the definition). For example, the 0.5-quantile is the median. If $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ are the order statistics corresponding to the X_j's from n independent replications, then a point estimator for x_q is the sample q-quantile $\hat{x}_q$, which is given by

$$\hat{x}_q = \begin{cases} X_{(nq)} & \text{if } nq \text{ is an integer} \\ X_{(\lfloor nq+1 \rfloor)} & \text{otherwise} \end{cases}$$

Let r and s be positive integers that satisfy $1 \leq r < s \leq n$. If n is "sufficiently large," then a 100$(1-\alpha)$ percent confidence interval for x_q is given by [see Conover (1999, pp. 143–148)]

$$P(X_{(r)} \leq x_q \leq X_{(s)}) \geq 1 - \alpha$$

where

$$r = \lceil nq + z_{\alpha/2}\sqrt{nq(1-q)} \rceil$$

and

$$s = \lceil nq + z_{1-\alpha/2}\sqrt{nq(1-q)} \rceil$$

EXAMPLE 9.22. For the bank of Example 9.14, suppose that we would like to decide how large a lobby is needed to accommodate customers waiting in the queue. If we let X be the maximum queue length as defined in Example 9.21, then we might want to build a lobby large enough to hold $x_{0.95}$ customers, the 0.95-quantile of X. From the 100 replications in the previous example, we obtained $\hat{x}_{0.95} = X_{(95)} = 20$. Thus, if the lobby has room for 20 customers waiting in queue, this will be sufficient for approximately 95 out of every 100 days. Furthermore, an approximate 90 percent confidence interval for $x_{0.95}$ is $[X_{(91)}, X_{(99)}] = [19, 23]$. (For this problem, X is a discrete random variable, so that the confidence level is approximate.)

Analysis for steady-state parameters

Suppose that we want to estimate the steady-state mean $\nu = E(Y)$, which is also generally defined by

$$\nu = \lim_{i \to \infty} E(Y_i)$$

often suggested for dealing with this problem is called *warming up the model* or *initial-data deletion*. The idea is to delete some number of observations from the beginning of a run and to use only the remaining observations to estimate ν. For example, given the observations $Y_1, Y_2, \ldots, Y_m$, it is often suggested to use

$$\bar{Y}(m, l) = \frac{\sum_{i=l+1}^{m} Y_i}{m - l}$$

$(1 \le l \le m - 1)$ rather than $\bar{Y}(m)$ as an estimator of ν. In general, one would expect

Welch's Procedure

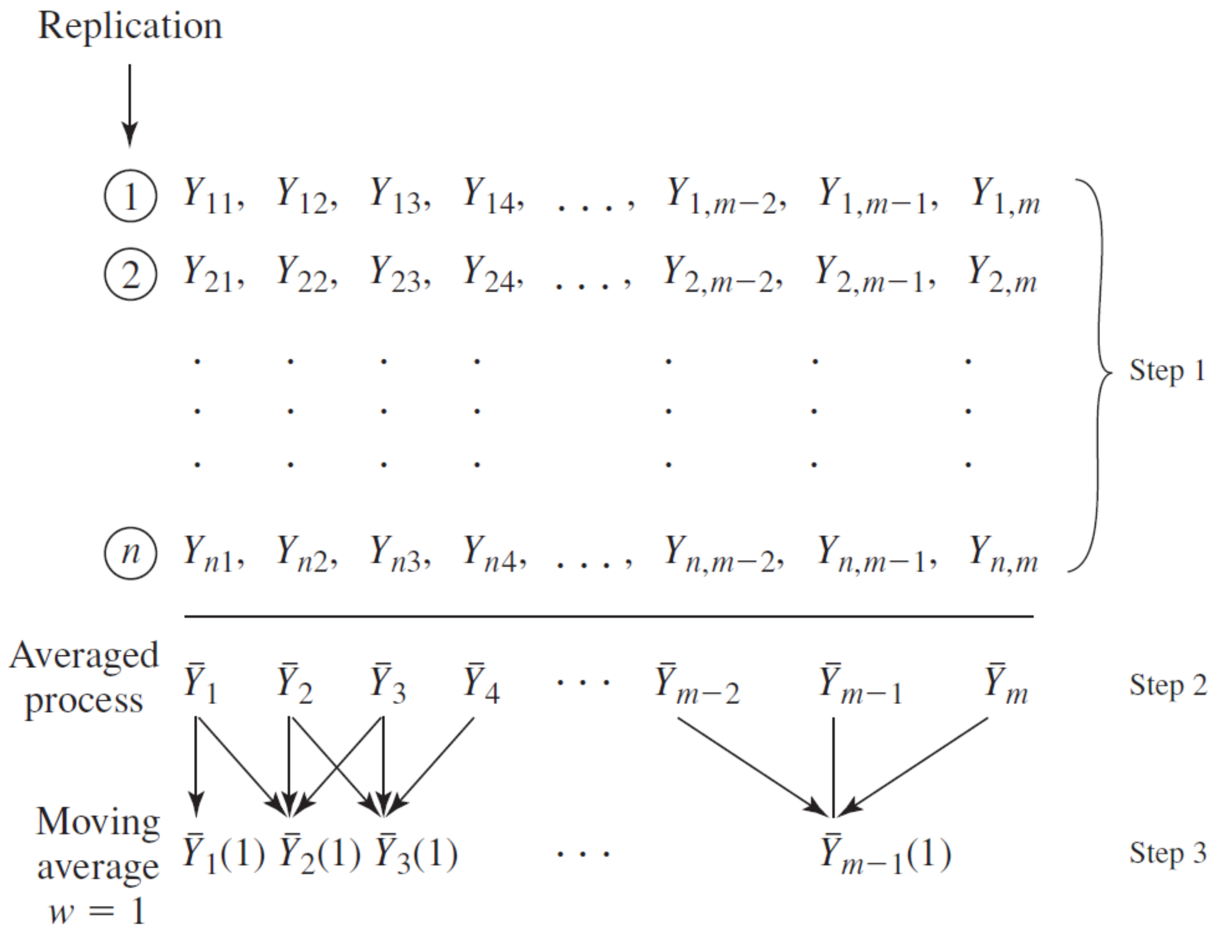

FIGURE 9.8
Averaged process and moving average with $w = 1$ based on n replications of length m.

Four Steps of Welch's Procedure

1. Make n replications of the simulation ($n \geq 5$), each of length m (where m is large). Let Y_{ji} be the ith observation from the jth replication ($j = 1, 2, \ldots, n$; $i = 1, 2, \ldots, m$), as shown in Fig. 9.8.
2. Let $\overline{Y}_i = \sum_{j=1}^{n} Y_{ji}/n$ for $i = 1, 2, \ldots, m$ (see Fig. 9.8). The averaged process $\overline{Y}_1, \overline{Y}_2, \ldots$ has means $E(\overline{Y}_i) = E(Y_i)$ and variances $\mathrm{Var}(\overline{Y}_i) = \mathrm{Var}(Y_i)/n$ (see Prob. 9.12). Thus, the averaged process has the same transient mean curve as the original process, but its plot has only $(1/n)$th the variance.
3. To smooth out the high-frequency oscillations in $\overline{Y}_1, \overline{Y}_2, \ldots$ (but leave the low-frequency oscillations or long-run trend of interest), we further define the *moving average* $\overline{Y}_i(w)$ (where w is the *window* and is a positive integer such that $w \leq \lfloor m/4 \rfloor$) as follows:

$$\overline{Y}_i(w) = \begin{cases} \dfrac{\sum_{s=-w}^{w} \overline{Y}_{i+s}}{2w+1} & \text{if } i = w+1, \ldots, m-w \\[2ex] \dfrac{\sum_{s=-(i-1)}^{i-1} \overline{Y}_{i+s}}{2i-1} & \text{if } i = 1, \ldots, w \end{cases}$$

4. Plot $\overline{Y}_i(w)$ for $i = 1, 2, \ldots, m - w$ and choose l to be that value of i beyond which $\overline{Y}_1(w), \overline{Y}_2(w), \ldots$ appears to have converged. See Welch (1983, p. 292) for an aid in determining convergence.

EXAMPLE 9.24. For simplicity, assume that $m = 10$, $w = 2$, $\overline{Y}_i = i$ for $i = 1, 2, \ldots, 5$, and $\overline{Y}_i = 6$ for $i = 6, 7, \ldots, 10$. Then

$$\overline{Y}_1(2) = 1 \qquad \overline{Y}_2(2) = 2 \qquad \overline{Y}_3(2) = 3$$
$$\overline{Y}_4(2) = 4 \qquad \overline{Y}_5(2) = 4.8 \qquad \overline{Y}_6(2) = 5.4$$
$$\overline{Y}_7(2) = 5.8 \qquad \overline{Y}_8(2) = 6$$

Before giving examples of applying Welch's procedure to actual stochastic models, we make the following recommendations on choosing the parameters n, m, and w:

- Initially, make $n = 5$ or 10 replications (depending on model execution time), with m as large as practical. In particular, m should be much larger than the anticipated value of l (see Sec. 9.5.2) and also large enough to allow infrequent events (e.g., machine breakdowns) to occur a reasonable number of times.
- Plot $\overline{Y}_i(w)$ for several values of the window w and choose the smallest value of w (if any) for which the corresponding plot is "reasonably smooth." Use this plot to determine the length of the warmup period l. [Choosing w is like choosing the interval width Δb for a histogram (see Sec. 6.4.2). If w is too small, the plot of $\overline{Y}_i(w)$ will be "ragged." If w is too large, then the $\overline{Y}_i$ observations will be overaggregated and we will not have a good idea of the shape of the transient mean curve, $E(Y_i)$ for $i = 1, 2, \ldots$.]
- If no value of w in step 3 is satisfactory, make 5 or 10 additional replications of length m. Repeat step 2 using all available replications. [For a fixed value of w, the plot of $\overline{Y}_i(w)$ will get "smoother" as the number of replications increases. Why?]

EXAMPLE 9.25. A small factory consists of a machining center and inspection station in series, as shown in Fig. 9.9. Unfinished parts arrive to the factory with exponential interarrival times having a mean of 1 minute. Processing times at the machine are uniform on the interval [0.65, 0.70] minute, and subsequent inspection times at the inspection

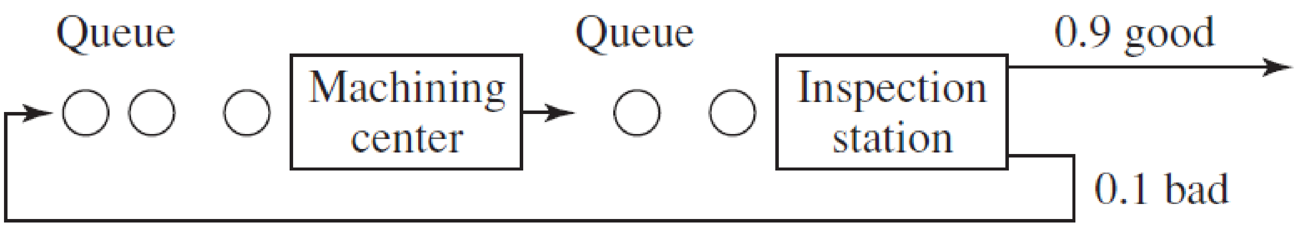

FIGURE 9.9
Small factory consisting of a machining center and an inspection station.

station are uniform on the interval [0.75, 0.80] minute. Ninety percent of inspected parts are "good" and are sent to shipping; 10 percent of the parts are "bad" and are sent back to the machine for rework. (Both queues are assumed to have infinite capacity.) The

Consider the stochastic process $N_1, N_2, \ldots$, where N_i is the number of parts produced in the ith hour. Suppose that we want to determine the warmup period l so that we

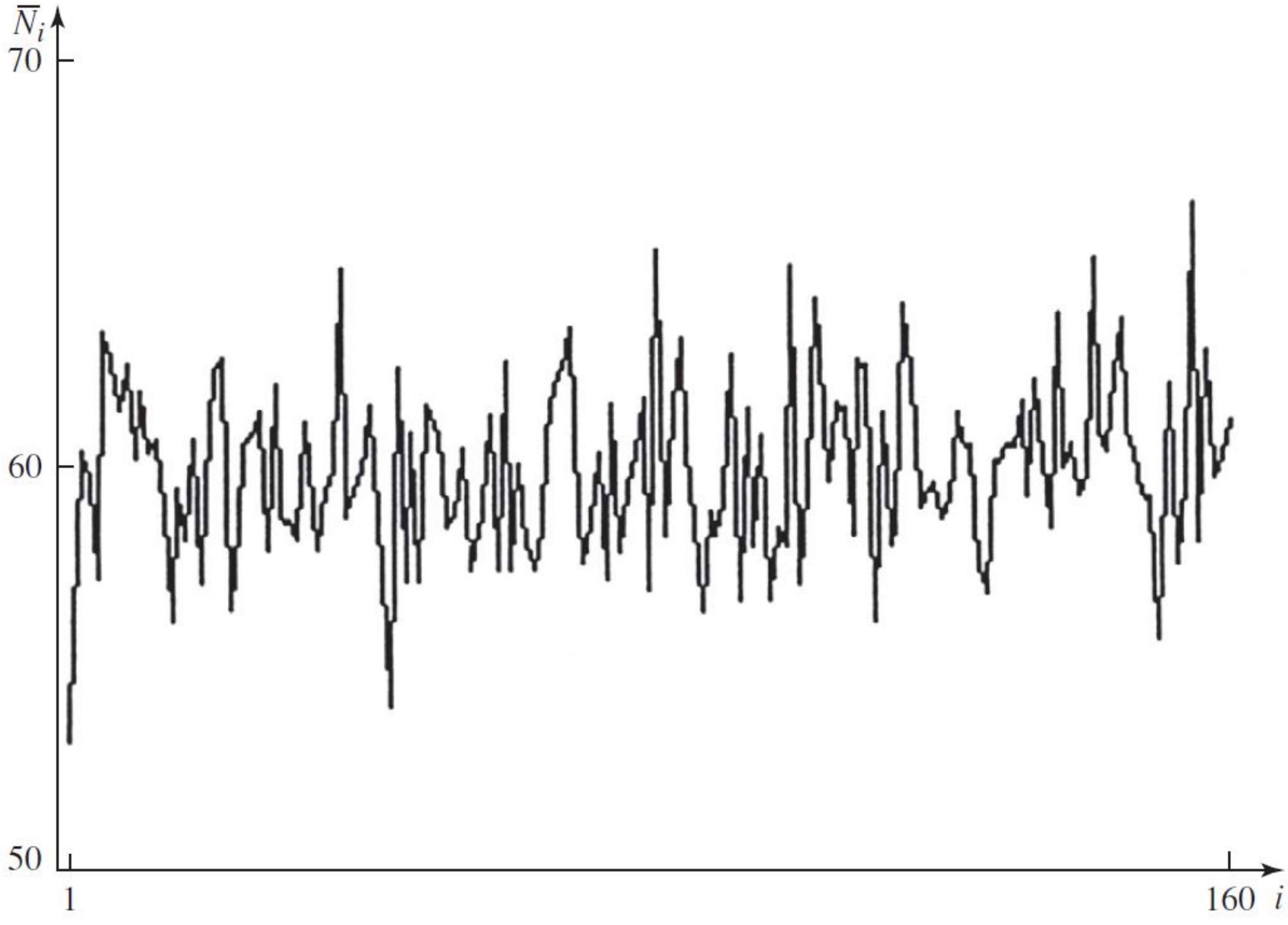

FIGURE 9.10
Averaged process for hourly throughputs, small factory.

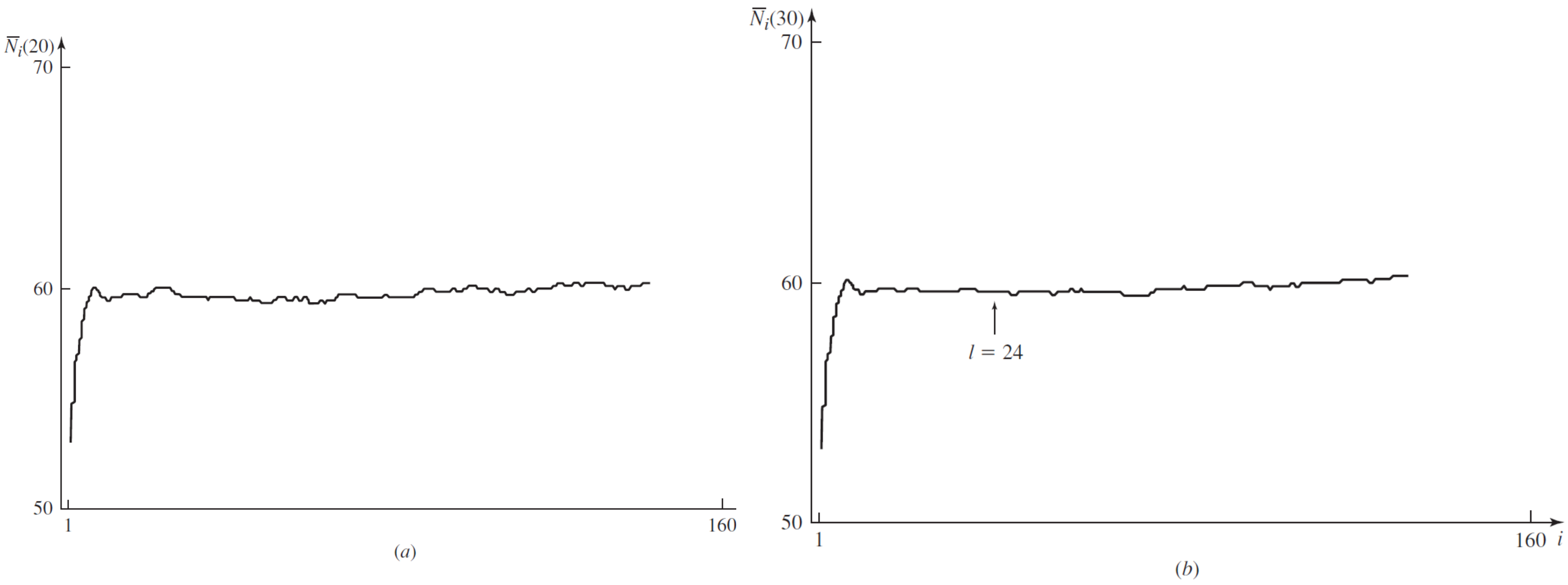

FIGURE 9.11
Moving averages for hourly throughputs, small factory: (*a*) $w = 20$; (*b*) $w = 30$.

Replication/Deletion Approach

We now present the *replication/deletion approach* for obtaining a point estimate and confidence interval for ν. The analysis is similar to that for terminating

n' replications of the simulation each of length m' observations, where m' is much larger than the warmup period l determined by Welch's graphical method (see Sec. 9.5.1). Let Y_{ji} be as defined before and let X_j be given by

$$X_j = \frac{\sum_{i=l+1}^{m'} Y_{ji}}{m' - l} \qquad \text{for } j = 1, 2, \ldots, n'$$

to "steady state," namely, $Y_{j,l+1}, Y_{j,l+2}, \ldots, Y_{j,m'}$.) Then the X_j's are IID random variables with $E(X_j) \approx \nu$ (see Prob. 9.15), $\overline{X}(n')$ is an approximately unbiased point estimator for ν, and an approximate $100(1 - \alpha)$ percent confidence interval for ν is given by

$$\overline{X}(n') \pm t_{n'-1,1-\alpha/2}\sqrt{\frac{S^2(n')}{n'}} \tag{9.6}$$

where $\overline{X}(n')$ and $S^2(n')$ are computed from Eqs. (4.3) and (4.4), respectively.